Professional Self-Assessment

Andrew Allen

Department of Computer Science, Southern New Hampshire University

CS 499: Computer Science Capstone

Dr. Sherin Aly

August 12th, 2026

Professional Self-Assessment

Completing my Bachelor of Science in Computer Science, building on my associate degree in information technology, has transformed my understanding of software development from primarily learning concepts to understanding how everything works together to creating real software solutions. Throughout the program, I developed skills in software engineering, algorithms and data structures, databases, security, testing, system analysis, and application development. More importantly the program taught me to approach software development as a process of analyzing requirements, selecting appropriate technologies, designing maintainable solutions, testing implementations, and communicating technical decisions to others.

My primary professional goal is to pursue a career in software engineering. The coursework and capstone experience have prepared me for this by giving me experience working across multiple layers of an application rather than focusing exclusively on writing code. I have learned to consider architecture, performance, persistence, security, usability, maintainability, and testing as interconnected parts of software quality. Building my ePortfolio has also helped me recognize how these skills complement one another and have given me a way to demonstrate my capabilities through concrete examples rather than simply describing them on a resume.

Professional Skills and Computer Science Foundation

Courses such as CS 250: Software Development Lifecycle, and CS 255: System Analysis and Design, taught me that successful software development requires more than writing code. I learned the importance of requirements analysis, documentation, development processes, version control, and communicating design decisions. My capstone project further strengthened this perspective by requiring me to evaluate and enhance an existing past project while considering how architectural changes affect the system as whole.

My coursework also strengthened my ability to translate technical requirements into understandable designs and explanations. Through system analysis, documentation, presentations, and project assignments, I learned to communicate not only what a system does, but why a particular decision was made and how it affects end users and stakeholders. This ability will be essential when working with others in a professional software environment.

During my CS 300: Data Structures and Algorithms course, I developed an ability to evaluate solutions based on computational efficiency, data organization, and complexity. Additionally, my CS 330: Computer Graphics and Visualization and CS 370: Current and Emerging Trends courses expanded on this foundation through computational graphics and artificial intelligence, requiring me to apply mathematical and algorithmic concepts to more complex problems. These experiences taught me to consider performance and scalability rather than simply producing a solution that just works.

Courses such as DAD 220: Intro to Structured Databases, CS 340: Client/Server Development, and CS 360: Mobile Architecture and Programming, gave me experience with database design, SQL, NoSQL, client/server development, data persistence, Python, Java, and various architecture designs. I learned the importance of separating responsibilities within an application and applying patterns such as Model-View-Controller (MVC), Model-View-

ViewModel (MVVM), and repository-based data access. These concepts became especially valuable during my capstone, where I applied architectural and database principles to improve my mobile inventory tracking application.

During my CS 305: Software Security and CS 320: Software Testing and Automation courses, I learned to treat security and reliability as fundamental parts of software development. I gained experience with secure coding practices, input validation, vulnerability awareness, and systematic testing. These courses reinforced the importance of designing software that not only performs its intended functions but also protects data and behaves predictably under unexpected conditions.

Portfolio Overview and Artifact Integration

The artifacts in this ePortfolio demonstrate how I progressed from learning individual computer science concepts to applying them within a complete software engineering project. The central artifact is my mobile inventory tracking application, originally developed during my CS 360 course and enhanced across all three areas.

The Software Engineering and Design enhancement demonstrates improvements to the applications architecture using Kotlin, MVVM, and modern Android navigation. The Algorithms and Data Structures enhancement demonstrates practical applications of multifield sorting and linear inventory calculations. The Database enhancement demonstrates persistent data management using Room and SQLite while maintaining separation between the database, repository, ViewModel, and user interface.

Together, these artifacts demonstrate many areas of my technical ability. Software engineering demonstrates my ability to structure and maintain an application, algorithms

demonstrate how I solve computational problems efficiently, and database development demonstrates how I organize, store, and retrieve information securely and reliably. Supporting coursework in areas such as system analysis, client/server development, security, testing, artificial intelligence, and computational graphics further demonstrate the breadth of my computer science journey.

Ultimately, this ePortfolio represents both what I have learned and my ability to apply that knowledge to practical software development. The Computer Science program has given me the technical foundation, problem solving skills, and professional perspective necessary to pursue my goal of becoming a software engineer and to continue learning and growing within the field.